

A MATHEMATICAL THEORY OF REUSABLE NEURAL BASES FOR NETWORK COMPRESSION

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ABSTRACT

Memory cost remains a critical bottleneck in both the training and deployment of large-scale deep AI models. Inspired by the trade-off between time and memory complexity in many classical algorithms, we introduce a novel RNN-based framework, the Linear Reusable Neural Bases Architecture (LRNBA), designed to improve parameter efficiency through parameter reuse *at the neuron level*. By introducing the concept of *neural bases*, our framework allows each neural network module to be represented as a linear combination of these neural bases with allowable shifts, enabling the same neurons to be reused across depths. This reuse mechanism reduces parameter redundancy across depths and improves the model compression rate, allowing for the construction of substantially wider and deeper networks under the same parameter budget. Extensive experiments show that our architecture achieves comparable or lower final loss than classical architectures, while maintaining stable training dynamics.

1 INTRODUCTION

Large-scale deep AI models outperform smaller ones across a broad range of domains (Brown et al., 2020; Hoffmann et al., 2022) and have attracted significant attention in the machine learning community. As indicated by empirical scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022), models with a larger parameter count typically have stronger approximation capabilities. Consequently, modern AI models are often designed with billions or even trillions of parameters, requiring hundreds or thousands of GPUs/TPUs for training and deployment (Brown et al., 2020; Fedus et al., 2022).

Despite large-scale deep AI models have demonstrated remarkable performance, training and deployment remain challenging, especially for researchers and organizations with limited computational resources. One of the major bottlenecks is the memory requirement. Training or deploying a large-scale model often requires operating hundreds or thousands of GPUs simultaneously, and the high cost of specialized hardware further increases the financial burden (Narayanan et al., 2021). Overall, these resource and hardware demands hinder many researchers and organizations from developing and deploying large-scale deep AI models.

A promising approach to addressing this challenge is to transform memory complexity into time complexity through a parameter-reuse mechanism, thereby reducing parameter redundancy and lowering memory costs. This idea is closely related to recurrent neural network (RNN) models, which repeatedly apply a shared transformation across different time steps or positions (Elman, 1990; Jordan, 1997). Furthermore, consistent with the philosophy that *compression leads to intelligence*, models that compress information using fewer parameters often exhibit better generalization. Nevertheless, classical RNN models often suffer from optimization difficulties, as repeated transformations can result in unstable gradient propagation, such as vanishing or exploding gradients. As a result, most large-scale deep AI models are designed with independent modules and integrate only limited reuse mechanisms.

In light of these observations, a natural question arises:

Can we design an architecture that integrates the parameter-reuse mechanism from RNNs to improve parameter efficiency and lower memory costs, while maintaining training stability?

In this paper, we introduce a novel model architecture, the Linear Reusable Neural Bases Architecture (LRNBA), to improve parameter efficiency through parameter reuse *at the neuron level*. Inspired by sinusoidal bases in Fourier analysis and linear bases in linear algebra, we introduce the concept of *neural bases*. We represent each neural network module as a linear combination of these neural bases with allowable shifts, enabling the same neurons to be reused across depths. This reuse mechanism reduces the parameter redundancy along the depth dimension and improves the model compression rate, allowing for the construction of substantially wider and deeper networks under the same parameter budget. Experimental results show that our architecture achieves comparable or lower final loss than classical architectures, while maintaining stable training dynamics and substantially reducing the parameter count.

Organization. The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 provides background on deep learning theory and residual networks. Section 4 introduces the theory of parameter reuse, the model architecture, and the training methods. Section 5 discusses the advantages and limitations of the model. Section 6 presents computational experiments to evaluate the model.

Notation. We use $\sigma(\cdot)$ to denote a nonlinear activation function, $\phi(\cdot)$ a neural basis function, and $R(\cdot)$ a residual block. The notation $\langle \cdot, \cdot \rangle$ denotes the inner product, and α denotes a scalar coefficient. We write d for the embedding dimension, m for the network width (i.e., the hidden dimension or number of neurons), L for the network depth, H for the number of attention heads, and T for the token sequence length. For a residual feedforward block, (S, b) denotes the sensor parameters, and P denotes the response parameters. For the j -th neural basis function (or neuron), the corresponding sensor and response parameters are denoted by (s_j, b_j) and p_j , respectively.

2 RELATED WORK

Several lines of research are related to our work on parameter reuse. We organize these studies into two main categories: recurrent neural networks (RNNs) and parameter-sharing methods beyond recurrent architectures.

Recurrent Neural Networks. Recurrent neural networks (RNNs) represent one of the most widely adopted forms of parameter sharing and reuse in neural architectures. Classical RNN models include the Elman network, the Jordan network, long short-term memory networks (LSTMs), and gated recurrent units (GRUs) (Elman, 1990; Jordan, 1997; Hochreiter & Schmidhuber, 1997; Cho et al., 2014). More recent RNN models improve parameter efficiency via low-rank factorizations and shared low-rank components (Lu et al., 2016), or leverage auxiliary networks, such as HyperNEAT, to generate related parameters across layers or time steps (Ha et al., 2016). Rebuffi et al. (2017); Savarese & Maire (2019) using learned linear combinations of shared parameter templates to reduce the parameter redundancy of CNN models and improve the arcitecture bias towards the recurrent networks.

Wang et al. (2025) also share a similar the spirit as ours, but their implementation is related on the indiidual layers, rather than neuron-based reuse.

Although RNN models are useful at reducing memory cost, standard RNNs remain difficult to optimize, as repeated multiplication through time can cause vanishing or exploding gradients during backpropagation through time (Hochreiter & Schmidhuber, 1997; Bengio et al., 1994; Pascanu et al., 2013). Classical LSTMs and GRUs alleviate this issue through gating controls. Other approaches improve stability via orthogonal recurrent transformations (Helfrich et al., 2018), residual or convolutional recurrent structures (Bai et al., 2019), and residual scaling techniques such as ReZero (Bachlechner et al., 2021).

Parameter Sharing Beyond RNNs. Parameter reuse has also been explored beyond recurrent models. Tensorized neural networks compress dense transformations using tensor-train decomposi-

tions (Novikov et al., 2015), while related methods employ structured matrices, such as Toeplitz-like, Hankel-like, and block-circulant matrices, to reduce storage costs (Sindhwani et al., 2015).

Parameter sharing is also widely studied in Transformers (Vaswani et al., 2017). The Universal Transformer/Relaxed Recursive Transformer repeatedly applies shared self-attention and transition blocks across depth (Dehghani et al., 2019; Bae et al., 2025). Pires et al. (2023) examine redundancy among Transformer feed-forward network (FFN) blocks and show that a single wide FFN module can be shared across multiple layers. The Sparse Universal Transformer further combines depth-wise sharing with sparse mixture-of-experts layers to increase capacity without activating all parameters at every iteration (Tan et al., 2023). ALBERT adopts cross-layer parameter sharing among Transformer encoders, reducing the number of independently learned parameters while maintaining competitive performance (Lan et al., 2020).

More recently, looped Transformers (Giannou et al., 2023; Yang et al., 2024) repeatedly apply a fixed block, or a small set of blocks, to an evolving hidden representation. Such models can emulate iterative computational procedures (Giannou et al., 2023) and provide a useful inductive bias for learning iterative algorithms and in-context tasks with fewer parameters (Yang et al., 2024).

Connect to Our Work. Our model can be regarded as an RNN-like architecture, as it reuses the same group of neurons across multiple modules. Unlike classical RNNs, which repeatedly apply an identical transformation, and other models that perform reuse at the *module* level, our method directly reuses nonlinear features at the *neuron level* across depths. This enables more diverse combinations and improved parameter efficiency, yielding better generalization while maintaining stable training.

3 BACKGROUND

Before presenting our main results, we briefly review several key concepts and formulations.

3.1 DEEP LEARNING

Deep learning theory suggests that the approximation ability of neural networks can be improved by increasing the depth of sequentially composed architectures (LeCun et al., 2015). By hierarchically composing multiple nonlinear transformations, deep neural networks can represent certain functions more efficiently than the shallow networks (Raghu et al., 2017). Consequently, deep networks often exhibit better approximation ability than the shallow networks for complex learning tasks.

Although the sequentially composed architecture of deep neural networks enables models to learn more complex features and representations, it also introduces substantially higher computational and memory costs during training or inference, compared to shallow networks (Rajbhandari et al., 2020). Since each module must be processed sequentially, a large number of parameters and intermediate activations must be held in memory at all times, even though only a small group of parameters are actively executing during training or inference. This memory retaining grows with network depth, posing a significant challenge for researchers with limited computational resources.

3.2 RESIDUAL NETWORKS

Residual networks (ResNets) play an important role in the deep learning area, as they alleviate the vanishing-gradient problem and enable the stable training of very deep networks (He et al., 2015). The key feature of a ResNet is the use of shortcut connections, which allows the input information to be passed directly to deeper layers rather than propagating only through nonlinear layers.

For example, a residual feedforward module in a ResNet can be written as

$$x + R(x),$$

where x represents the shortcut connection and $R(x)$ represents the residual component. A typical residual component takes the form

$$R(x) = P\sigma(Sx + b),$$

where $\sigma(\cdot)$ is an element-wise nonlinear activation function, and (S, b) are the learnable parameters of the input affine layer, and P are the learnable parameters of the output linear layer.¹

From a more detailed perspective, the residual component can be viewed as a linear combination of a shared collection of nonlinear functions. The parameters (S, b) determine the directions and intercepts (offsets) of these nonlinear functions in the input space, while P specifies the coefficients used to combine them in the output space. We further extend this viewpoint in Section 4 to develop a new model architecture that reuses parameters to reduce memory costs.

4 METHODOLOGY

In this section, we introduce our novel architecture, the Linear Reusable Neural Bases Architecture (LRNBA), designed to improve parameter efficiency and reduce memory cost. The key idea is to reuse a shared group of neural bases while assigning different coefficients to construct blocks at different positions, enabling the same neuron to be reused multiple times. In the remainder of this section, we first define neural bases and provide a geometric interpretation from the perspective of vector fields. We then describe the model architecture, training strategy, and key implementation details.

4.1 THE NEURAL BASES

In this subsection, we first introduce our core concept of neural bases and provide a geometric interpretation from the perspective of vector fields.

Definition of Neural Bases. As introduced in section 3, a residual feedforward block has the form

$$x + P\sigma(Sx + b). \quad (1)$$

Let m denote the hidden dimension of the residual block, and write the rows of S , the entries of b , and the columns of P as

$$S^\top = [s_1, s_2, \dots, s_m], \quad b^\top = [b_1, b_2, \dots, b_m], \quad P = [p_1, p_2, \dots, p_m],$$

where s_j is the j -th row of S , b_j is the j -th entry of b , and p_j is the j -th column of P .

For each $j = 1, 2, \dots, m$, we define the function $\phi_j(x)$ as:

$$\phi_j(x) := \sigma(\langle x, s_j \rangle + b_j)p_j.$$

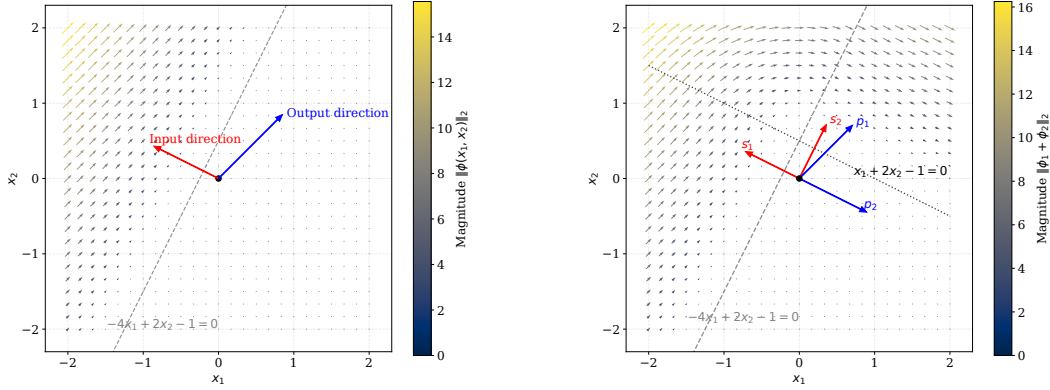
Then the residual component of the network can be rewritten as a summation of such functions:

$$P\sigma(Sx + b) = \sum_{j=1}^m \sigma(\langle x, s_j \rangle + b_j)p_j = \sum_{j=1}^m \phi_j(x). \quad (2)$$

This formulation reveals that the mapping of the residual block can be further decomposed into a group of basic nonlinear functions $\{\phi_j\}_{j=1}^m$. Based on this decomposition, and since each of these functions serve as an *atomic* nonlinear transformation from the input to the output space—in the sense of being the simplest and independent building block—we refer to each function ϕ_j as a *neuron* or *neural basis* of the network, analogous to a sinusoidal basis in Fourier analysis or a basis vector in linear algebra. Moreover, we refer to the associated parameters (s_j, b_j) for the input space as the *sensor* of neuron ϕ_j and p_j as the *response* of neuron ϕ_j .

Superposition of Vector Fields. Here, we provide a geometric interpretation of the neural basis. Specifically, each neural basis can be viewed as a vector field from the input space to the output space. For the j -th neuron, as illustrated in Figure 1a, the sensor vector s_j and scalar b_j specifies the direction and bias of the input vector field, and vector p_j determine the response vector in the output vector field. These parameters are connected through a bilinear structure between the nonlinear activation $\sigma(\langle x, s_j \rangle + b_j)$ and the response vector p_j . As a result, a residual feedforward block can be viewed as a superposition of many such vector fields, as depicted in Figure 1b.

¹For simplicity, we omit the bias term in the output layer. This omission does not affect the nonlinear structure of the residual component, since the output bias contributes only a constant translation.



(a) Visualization of the vector field defined by a single neural basis.

(b) Visualization of a superposition of two vector fields induced by two neural bases.

Figure 1: Geometric interpretation of neural bases from the perspective of vector fields. **(a)** Visualization of a single neural basis, defined by the function $\phi(x) = \sigma(\langle x, s \rangle + b)p$ with parameters $s = (-4, 2)$, $b = -1$, and $p = (1, 1)$. The red and blue arrows indicate the directions of sensor s and response p in the input and output vector fields, respectively. The dashed line marks the decision boundary induced by the equation $\langle x, s \rangle + b = 0$. The small arrows on the grid illustrate the push-forward effect of the vector field. **(b)** A superposition of two vector fields, where the first neural basis is the same as that shown in **(a)** and the second basis has parameters $s_2 = (1, 2)$, $b_2 = -1$, and $p_2 = (2, -1)$. Note that a single neuron can only cause one directional push-forward, whereas the combination of two neurons can generate multiple directional effects, as exhibited by the rotational dynamics in **(b)**.

Projection Neural Bases. Beyond residual feedforward blocks, the concept of neural bases is also extendable to other types of neural network components, such as projection neural bases. Due to space limitations, detailed formulations are provided in Appendix A.

The above generalized view enables us to consistently treat different types of network modules as linear combinations of corresponding neural bases, extending the idea of reusable neural bases to a broader range of neural network architectures, including transformers.

4.2 LINEAR REUSABLE NEURAL BASES ARCHITECTURE

We first introduce two core principles guiding our architecture design: *neuron reuse* and *linear construction*. We then describe the two major components of our model following from these principles: the reusable neural bases and the coefficient table. Finally, we discuss the training strategy, implementation details, and further improvement techniques, including sub-neuron reuse and parameter-sharing methods.

Throughout this section, we consider a standard residual network comprising L stacked feedforward blocks.

Model Architecture. Accordingly, our proposed architecture consists of two components:

1. **A collection of reusable neural bases**, denoted by $\{\phi_j\}_{j=1}^m$, which serve as function bases for constructing different blocks.
2. **A coefficient table**, denoted by $\{\alpha_{j,l}\}_{j=1,l=1}^{m,L}$, containing scalar values that determine the contribution of each neural basis to each block.

Specifically, the residual block $R_l(x)$ at depth l is constructed via a linear combination of the shared bases $\{\phi_j\}_{j=1}^m$ with position-wise coefficients $\{\alpha_{j,l}\}_{j=1}^m$:

$$R_l(x) = \sum_{j=1}^m \alpha_{j,l} \phi_j(x).$$

Under this formulation, the same collection of neural bases can be reused across all depths, while each block at a different position remains distinct due to its different coefficients.

Note that here we assume each real-value coefficient $\alpha_{j,l}$ is independent. In fact, one may impose constraints on the coefficients, such as non-negativity or binary values, influencing on the topological connectivity of the network and yielding sparse or dense structures. Due to page limitations, we leave these investigations for future work.

Algorithm 1 Average accumulated gradients of reused neural bases.

```

1: for each parameter  $p$  in reused layers do
2:   Attach a gradient hook to  $p$ 
3:   Define hook function:  $\text{grad}, \text{depth} \mapsto \text{grad}/\text{depth}$ 
4:   Register the hook to divide accumulated gradients by depth
5: end for
```

Algorithm 2 Forward pass across all depth blocks via shared neural bases.

```

1: for  $i \leftarrow 1$  to  $L$  do
2:    $a \leftarrow \text{reused\_sensors}(x)$ 
3:    $a \leftarrow \text{Activation}(a)$ 
4:    $a \leftarrow a \odot \mathbf{c\_table}[:, i]$ 
5:    $a \leftarrow \text{reused\_responses}(a)$ 
6:    $x \leftarrow x + a$ 
7: end for
```

Training Strategy and Implementation Details. Our training approach mostly follows the standard gradient descent framework. However, because the neural bases are reused across multiple blocks, their gradients are accumulated over all positions for each optimization step. To maintain the magnitude of gradient updates consistent with standard models, we rescale the accumulated gradients of the neural bases by the number of their reused times. In contrast, the parameters of the coefficient table are updated directly without rescaling. The implementation details are summarized in Algorithm 1 and 2.

For initialization, the sensor layers of the reusable neural bases are initialized using Kaiming initialization (the default PyTorch setting), while the response layers are zero-initialized to provide a neutral starting point for stable training, following the guidance of Bachlechner et al. (2021); Zhang et al. (2019). The coefficient table parameters are initialized from a standard Gaussian distribution $\mathcal{N}(0, 1)$.²

Sub-neural Level Reuse. Beyond sharing neural bases, parameter counts can be further reduced by reusing sub-neural structures—specifically, sensors or responses—at a finer granularity. One simple approach is to let multiple neurons share a single response vector, yielding a more complex neuron structure with multiple sensors and one response, which we refer to as a multi-sensor neuron. Networks containing Maxout (Goodfellow et al., 2013) or SwiGLU (Shazeer, 2020) activation functions can be viewed as examples built from such multi-sensor neurons.

These multi-sensor neurons not only can reduce the parameter counts but also alleviate the dead neuron issue commonly observed in single-sensor neurons. By merging multiple sensors, the active domain of the neuron is expanded across the input space, enabling more efficient feature representation and more stable training dynamics.

5 DISCUSSION

Compared to classical neural networks, the reuse principle in our model introduces several new characteristics, particularly in terms of architecture shape, network compression, and coefficient energy. We discuss these characteristics below, followed by a brief discussion of the limitations.

5.1 ARCHITECTURE SHAPE

In classical neural networks, the parameter count is directly tied to the architecture shape, depending on the embedding dimension, width (hidden dimension), and depth. Designing the network architec-

²Although other initialization schemes are possible, simultaneously zero-initializing both the coefficient table and the response vectors should be avoided. Since these two components are connected through a bilinear structure, zero initialization would cause their gradients to vanish entirely.

ture requires balancing the trade-offs among these dimensional factors and predefining the shape to achieve good performance. In contrast, our model reuses a shared collection of neural bases across all depths, making the total parameter count mainly dependent on width and almost decoupled from network depth. Although the coefficient table also grows linearly with depth L , its memory footprint remains negligible compared to the full-weight matrices of classical architectures. Consequently, much wider and deeper networks can be constructed under the same memory budget.

5.2 NETWORK COMPRESSION

Another interesting point is related to network compression. Unlike classical networks, where compression is primarily achieved at the data-representation level by approximating the target function with fewer parameters, our approach introduces compression on two additional levels:

1. **Component-Level Compression:** Reusing the same neural bases across depths allows the network to capture recurrent or cyclical patterns more effectively, yielding a more parameter-efficient representation.
2. **Topological Connection Compression:** Our model enables the construction of wider and deeper networks, increasing the opportunity to discover essential topological interconnections among neurons and thereby forming networks at a high compression rate.

As indicated by the philosophy that “compression is intelligence”, high compressed models with fewer parameters typically exhibit better robustness to noise and better generalization performance.

5.3 COEFFICIENT ENERGY AND PRUNING

Inspired by the spectral energy in Fourier analysis, we may define the energy of a neural basis via its coefficients. For instance, we can define the energy of the j -th neuron as the sum of its squared coefficients across all depths:

$$E(\phi_j) = \sum_{l=1}^L \alpha_{j,l}^2.$$

This energy reflects the importance or frequency of the j -th neural basis with respect to the data and the network architecture, and can serve as a criterion for pruning less important neurons to further reduce memory cost.

5.4 LIMITATIONS

A primary limitation of our method is the increased computational time and energy consumption compared to classical models, as block in our model must be dynamically computed and updated during backpropagation. Nevertheless, this computational cost is confined to the training phase; it can be eliminated during inference by precomputing and caching the effective weights for each block.

6 EXPERIMENTS

We evaluate the proposed model in two settings: an FFN-based ResNet for function approximation and a decoder-only Transformer for language models. All experiments are conducted on a Linux workstation equipped with an Intel Core i7 CPU, 16 GB of RAM, and an NVIDIA RTX 3090 GPU. For fair comparisons, all models use the same random seed and identical mini-batch sampling order throughout training.

6.1 EVALUATION ON AN FFN-BASED RESNET

We first evaluate the proposed model on a nonlinear regression task.

Dataset and Training Setup. The target function is the composite function $f(x) = \cos(\cos x)$, which provides a nonlinear benchmark for evaluating function-approximation ability. We uniformly sample $N = 60,000$ points from the interval $[-1, 1]$ and randomly divide them into training and test

sets using a 3:1 ratio. All models are trained for 2,048 epochs with a batch size of 512. We use the mean squared error (MSE) loss and the AdamW optimizer with a learning rate of $lr = 1 \times 10^{-5}$.

Model Configuration. The baseline architecture is an FFN-based residual network consisting of $L = 8$ stacked residual blocks. Two additional projection blocks are placed at the beginning and end of the network to map the input to the embedding space and the final representation to the output space, respectively. The embedding dimension is set to $d = 64$, and the hidden dimension, corresponding to the number of neurons in each FFN block, is set to $m = 256$. To improve training stability and ensure a fair comparison, the projection blocks use the sensor-response duplex structure, with the response layers initialized to zero.

Evaluation Metrics. We evaluate the models using training and test MSE, per-epoch training time, and visual comparisons between the learned predictions and the target function. Due to space limitations, the test MSE loss results and prediction visualizations are reported in Appendix A.2. Figure 2 presents the training-loss and training-time comparisons.

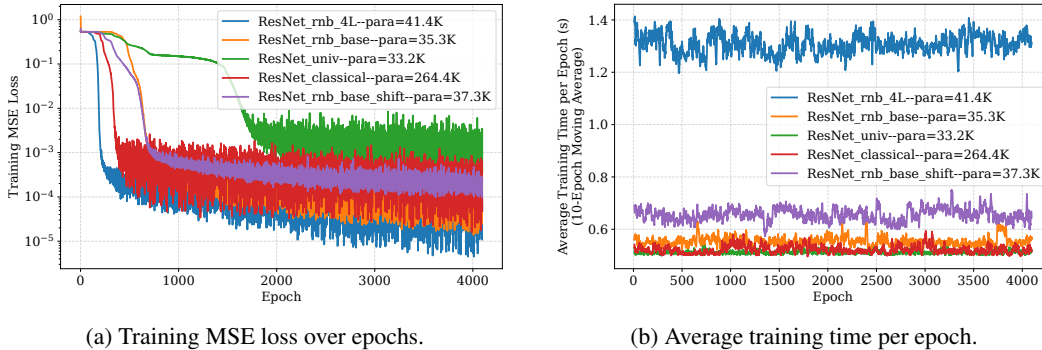


Figure 2: Comparison of FFN-based ResNet models on the nonlinear regression task. **(a)** Training MSE loss over epochs. The proposed LRNBA models achieve competitive or even faster convergence speed while using substantially fewer parameters than the standard ResNet baselines. **(b)** Smoothed training time over epochs. LRNBA introduces a modest training-time overhead because its residual blocks are dynamically constructed from reusable neural bases and depth-dependent coefficients.

Results. As shown in Figure 2(a), LRNBA achieves a convergence rate and final loss comparable to those of the standard ResNet while using only 1/7 of its parameters. It also substantially outperforms the parameter-matched *classical small* model and the baseline that repeatedly applies the same residual block across depth.

Increasing the width by a factor of four ($4m$) further accelerates convergence while achieving a comparable final loss with only half as many parameters as the standard ResNet. Increasing the depth by a factor of four ($4L$) yields even faster convergence and a lower final loss while using only 1/6 of the parameters. These results indicate that depth expansion provides greater performance gains than width expansion and that LRNBA improves parameter efficiency without compromising training stability or accuracy.

As shown in Figure 2(b), LRNBA incurs a modest increase in per-epoch training time because its residual blocks are dynamically constructed from reusable neural bases and their coefficients. This overhead can be removed during inference by precomputing and caching the effective weights.

6.2 EVALUATION OF THE TRANSFORMER MODEL

To further evaluate the proposed architecture, we apply it to a decoder-only Transformer on a language-modeling task.

Dataset and Tokenization. We conduct experiments on the WikiText-2 dataset. We use a byte-level BPE tokenizer with a total vocabulary size of $2^{14} = 16,384$, including 4 special tokens (e.g., UNK,

PAD, EOS, SOS). We first concatenate all sentences into a single sequence, and then randomly sample overlapping chunks of sequence length $T = 128$ tokens.

Model Configuration. For the language model, we use the standard Transformer (Vaswani et al., 2017) as the base model, with the following configuration: embedding dimension $d = 512$, depth $L = 8$, and $H = 8$ attention heads (head dimension 64). The hidden dimension of the FFN-based blocks is set to four times the embedding dimension, i.e., 2048, and no dropout is applied.

Training Setup. Each model is trained for 800 epochs using the AdamW optimizer with a learning rate of $lr = 1.5 \times 10^{-5}$. Each epoch consists of 512 training batches with a batch size of 16. For evaluation, we use 128 test batches. Although the original setup states that MSE loss is used for training, language modeling is typically optimized using cross-entropy loss; therefore, this should be verified for consistency.

Evaluation Metrics. We evaluate the models using training and test cross-entropy (CE) loss, per-epoch training time, and parameter count. Due to space limitations, the test MSE loss results and prediction visualizations are reported in Appendix A.2.

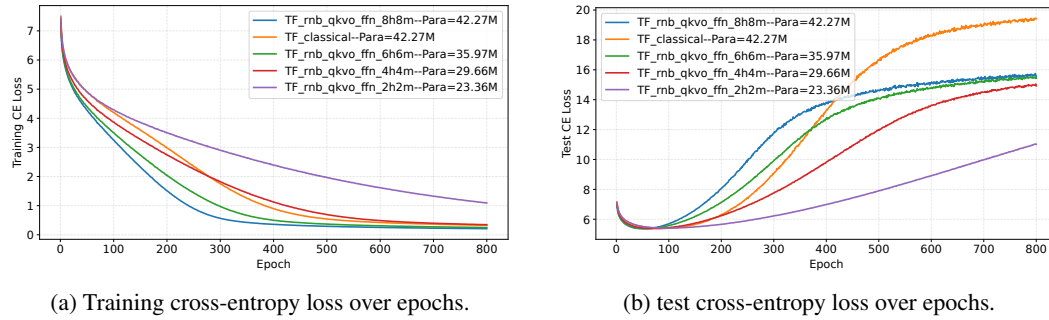


Figure 3: Comparison of decoder-only Transformer models on the WikiText-2 language-modeling task. **(a)** Training cross-entropy loss over epochs. Adding more reusable attention and FFN bases steadily improves convergence speed and lowers final training loss. The LRNBA model with expansion factor 6 (6h6m) for both heads and FFN width converges faster and achieves lower training loss than the standard Transformer, despite using only about 3/4 as many parameters. **(b)** Test cross-entropy loss over epochs. The classical Transformer yields the highest test CE loss, while the LRNBA models demonstrate that reducing parameter size consistently lowers test loss. This observation is consistent with the philosophy that compression is intelligence—the model with fewer parameters tends to achieve better generalization on test data.

Results. As shown in Figure 6.2(a), increasing the numbers of reusable attention and FFN bases consistently accelerates convergence and reduces the final training loss. In particular, The LRNBA model with expansion factor 6 (6h6m) for both heads and FFN width converges faster and achieves lower training loss than the standard Transformer, despite using only about 3/4 as many parameters. These results demonstrate that the proposed reuse mechanism improves parameter efficiency while maintaining stable training dynamics.

As shown in Figure 6.2(b), The classical Transformer yields the highest test CE loss, while the LRNBA models demonstrate that reducing parameter size consistently lowers test loss. This observation is consistent with the philosophy that compression is intelligence—the model with fewer parameters tends to achieve better generalization on test data.

Nevertheless, the proposed models remain substantially more parameter-efficient than the standard Transformer, making them well suited to memory-constrained applications where training time is a secondary consideration.

7 CONCLUSION

In this paper, we introduced the Linear Reusable Neural Bases Architecture (LRNBA) to improve parameter efficiency and reduce memory requirements. By decomposing feedforward neural networks into reusable neural bases and linear coefficients, we provided a geometric interpretation of neural computation through vector fields.

Guided by the principles of neuron reuse and linear construction, LRNBA model enables the construction of substantially wider and deeper networks under a limited memory budget. Our experiments on FFN-based ResNets and decoder-only Transformers demonstrate that LRNBA significantly reduces the parameter count while maintaining stable training and competitive performance. In several settings, expanding the number of reusable bases or network width and depth further improves convergence and final accuracy, highlighting the potential of neural-basis reuse for memory-constrained deep learning studies.

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A ATTENTION NEURAL BASES AND PROJECTION NEURAL BASES

A.1 ATTENTION NEURAL BASES

In this subsection, we provide a detailed description of attention neural bases.

A standard multi-head attention block with H attention heads can be written as

$$\text{Attn}(x) = \text{Concat}(A_1(x), A_2(x), \dots, A_H(x)) O,$$

where O denotes the output projection matrix and the output of the j th attention head is

$$A_j(x) = \text{softmax}\left(\frac{q_j k_j^\top}{\sqrt{d_H}}\right) v_j.$$

Here,

$$q_j = Q_j x, \quad k_j = K_j x, \quad v_j = V_j x,$$

are the query, key, and value representations associated with the j -th attention head, and d_H denotes the dimension of each head.

From the neural-basis perspective, each attention head A_j can be viewed as a nonlinear basis function that extracts a particular interaction pattern from the input sequence. Therefore, we refer to each attention head as an *attention neural basis* of the network, and the parameters (Q_j, K_j) are called the *sensor* of the attention neural basis A_j , while V_j is called the *response* of the attention neural bases. The complete attention output is then constructed by concatenating the outputs of these basis functions and combining them through the output projection O .

Similarly, we introduce scalar coefficients α_j for those heads. The resulting attention block is given by

$$\text{Attn}_\alpha(x) = \text{Concat}(\alpha_1 A_1(x), \alpha_2 A_2(x), \dots, \alpha_H A_H(x)) O,$$

where the coefficients $\{\alpha_j\}_{j=1}^H$ determine the relative coefficient of the attention heads to the final output.

A.2 PROJECTION NEURAL BASES

In this subsection, we provide a detailed description of projection neural bases, including both the derivation and visualizations.

Derivation Instead of representing each projection block as a single linear layer, we adopt a sensor–response duplex structure, using the identity function as the activation functions. This duplex structure still works as a linear projection, as it consists of two linear layers produced in sequence.

Specifically, for a composite projection block, we adopt the same notation as in the FFN-based ResNet for the sensor (S, b) and response P , and write it as

$$P(Sx + b).$$

Let m denote the hidden dimension of the block, and write the rows of S , the entries of b , and the columns of P as

$$S^\top = [s_1, s_2, \dots, s_m], \quad b^\top = [b_1, b_2, \dots, b_m], \quad P = [p_1, p_2, \dots, p_m],$$

where s_j is the j -th row of S , b_j is the j -th entry of b , and p_j is the j -th column of P .

For each $j = 1, 2, \dots, m$, we define the function $\phi_j(x)$ as:

$$\phi_j(x) := (\langle x, s_j \rangle + b_j) \cdot p_j,$$

where the activation function $\sigma(x)$ is replaced with identify functions.

The projection block can be rewritten as a summation of such functions:

$$P(Sx + b) = \sum_{j=1}^m (\langle x, s_j \rangle + b_j) p_j = \sum_{j=1}^m \phi_j(x). \quad (3)$$

This formulation reveals that the projection block can be further decomposed into a collection of more elementary functions $\{\phi_j\}_{j=1}^m$.

Based on this decomposition, we refer to each function ϕ_j as a *projection neuron* or *projection neural basis* of the network, and similarly, we call the parameters (s_j, b_j) the *sensor* of neuron ϕ_j and p_j the *response* of neuron ϕ_j .

An interesting observation is that each projection neural basis actually corresponds to a rank-one linear(affine) transformation from the input space to the output space, since the expression can be expanded as

$$\phi_j(x) = (\langle x, s_j \rangle + b_j) \cdot p_j = p_j s_j^\top x + p_j b_j,$$

where $p_j s_j^\top$ is a rank-one matrix and $p_j b_j$ serves as a bias of the linear transformation.

Visualizations We also provide a visualization of the projection neural basis via vector fields. For simplicity, we adopt the same parameter values as those used in the earlier visualizations (see Figure 4).

For the j -th projection neuron, as illustrated in Figure 4a, the sensor vector s_j and scalar b_j specifies the direction and bias of the input vector field, and vector p_j determine the response vector in the output vector field. These parameters are connected through a outer product between value $\langle x, s_j \rangle + b_j$ and the response vector p_j . Consequently, a residual feedforward block can be actually interpreted as a superposition of vector fields, as shown in Figure 4b.

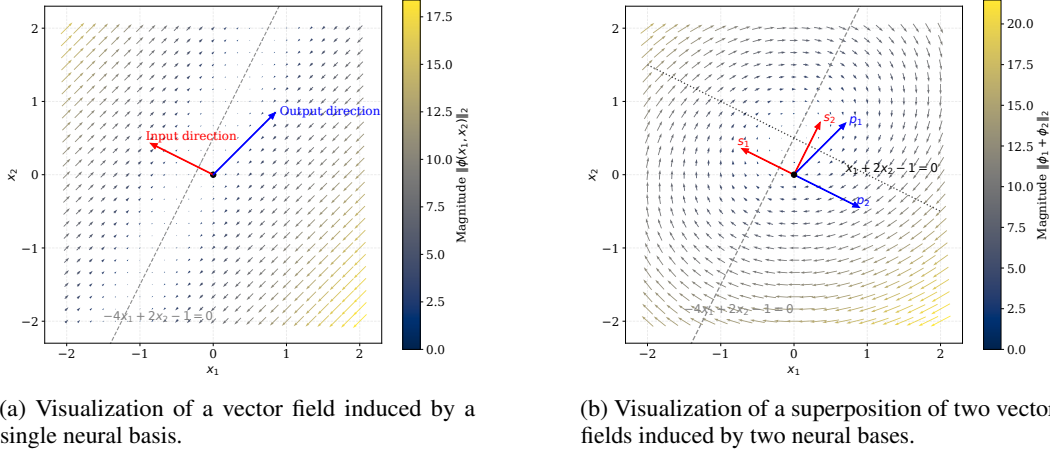


Figure 4: Geometric interpretation of projection neural bases from the perspective of vector fields. **(a)** Visualization of a single projection neural basis, defined by the function $\phi(x) = (\langle x, s \rangle + b)p$ with parameters $s = (-4, 2)$, $b = -1$, and $p = (1, 1)$. The red and blue arrows indicate the directions of the input and output vector fields, respectively, while the dashed line marks the decision boundary between the active and inactive domains. The small arrows on the grid illustrate the push-forward effect for the corresponding input positions induced by this basis function. **(b)** A superposition of two vector fields, where the first projection neural basis is identical to that shown in **(a)** and the second projection neural basis has parameters $s_2 = (1, 2)$, $b_2 = -1$, and $p_2 = (2, -1)$. Note that a single neuron induces only a directional vector effect, whereas the combination of two neurons can generate a field exhibiting rotational dynamics, as illustrated in **(b)**.

ADDITIONAL EXPERIMENTAL RESULTS OF FFN-BASED RESNET

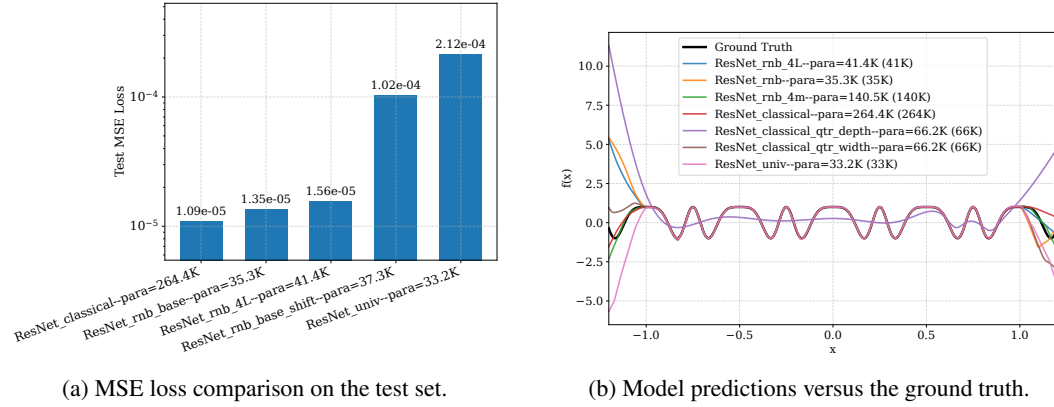


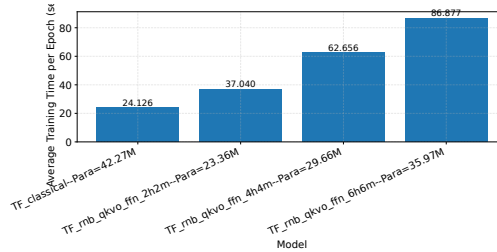
Figure 5: Test-set performance of LRNBA and classical ResNet variants. **(a)** Test MSE loss comparison; **(b)** Visualization of model predictions against ground truth.

Results. As shown in Figure 5(a), the test-set results are consistent with the training-set observations. The LRNBA model with a fourfold increase in depth achieves the lowest MSE, whereas the classical model with one-quarter of the original depth produces the highest loss. The LRNBA model also outperforms the baseline that directly reuses the same residual block.

These results support two main observations: depth expansion provides a larger reduction in test error than width expansion, and the proposed reuse mechanism improves parameter efficiency while preserving strong generalization performance.

As shown in Figure 5(b), all models except the one-quarter-depth classical variant closely approximate the ground-truth function. This comparison further highlights the importance of depth for accurate function approximation and demonstrates that LRNBA enables deeper networks under a limited parameter budget.

ADDITIONAL EXPERIMENTAL RESULTS OF TRANSFORMER MODELS



(a) Training cross-entropy loss over epochs.

Figure 6: Average training time per epoch. Because of parallel computation, the training and inference times are primarily determined by the network depth and are relatively insensitive to the network width. In practice, however, we observe that the computational time may also increase approximately with the number of neural bases. This growth may be related to the limited on-chip SRAM capacity of GPUs and could potentially be alleviated by larger on-chip memory or more efficient memory-management techniques.

Results. As shown in Figure 6, the reduction in memory usage comes with additional training-time overhead. Since the attention and FFN blocks are dynamically constructed from reusable neural bases

and their associated coefficients, the average training time increases approximately linearly with the number of bases.